

CASE STUDY

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Seasonal water quality assessment and GIS-based pollution mapping of the Brantas River toward sustainable water management and SDG 6

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Abstract

The Brantas River is an important freshwater resource supporting domestic, agricultural, industrial, and ecological functions in East Java, Indonesia. Increasing urbanization, industrial expansion, agricultural intensification, and population growth have contributed to substantial environmental pressures on river water quality, creating challenges for sustainable watershed management and Sustainable Development Goal 6 (SDG 6) implementation. This study evaluated seasonal water quality variation and developed GIS-based pollution mapping to identify the spatial distribution of pollution intensity along the Brantas River. Physicochemical parameters including pH, dissolved oxygen, biological oxygen demand, chemical oxygen demand, total suspended solids, nitrate, and phosphate were analyzed during dry and wet seasons at multiple monitoring stations. Water Quality Index analysis was applied to determine river water quality status, while Geographic Information System (GIS)-based spatial analysis was used to visualize pollution distribution and identify hotspot areas. The results showed clear seasonal differences in key water-quality variables. pH remained within the applicable range of 6–9, while nitrate concentrations remained below the 10 mg L⁻¹ limit in both seasons. In contrast, biological oxygen demand exceeded the 3 mg L⁻¹ standard during the dry season (6.85 ± 1.73 mg L⁻¹) and wet season (4.92 ± 1.41 mg L⁻¹). Total suspended solids also exceeded the 50 mg L⁻¹ standard, increasing from 68.42 ± 15.26 mg L⁻¹ in the dry season to 94.71 ± 20.38 mg L⁻¹ in the wet season. Phosphate concentrations exceeded the 0.2 mg L⁻¹ standard in both seasons, reaching 0.48 ± 0.11 mg L⁻¹ during the dry season and 0.73 ± 0.16 mg L⁻¹ during the wet season. The weighted arithmetic Water Quality Index ranged from 42 to 95 across the monitored river sections, indicating water-quality conditions from good to very poor. GIS-based mapping identified higher pollution intensity in urbanized and downstream river reaches, where cumulative domestic wastewater, industrial discharge, and agricultural runoff were most evident. These findings indicate that organic pollution during the dry season and sediment–nutrient loading during the wet season are the principal water-quality pressures affecting the Brantas River.



Priority actions should focus on domestic wastewater control, industrial effluent compliance, agricultural runoff reduction, and seasonally targeted monitoring to improve river-water quality and support SDG 6 implementation.

Keywords: Brantas River, Water quality index, GIS spatial analysis, Seasonal water quality, SDG 6

1 Introduction

River ecosystems support ecological balance, economic activity, and human well-being by providing water for domestic use, agriculture, industry, fisheries, energy production, and aquatic habitats. These multiple functions make river-water quality essential to freshwater security and ecosystem sustainability. Surface-water resources are increasingly exposed to environmental degradation due to rapid urbanization, industrial expansion, agricultural intensification, and population growth [1, 2]. Deterioration of river water quality has become a major environmental challenge because polluted rivers threaten aquatic ecosystems, reduce freshwater availability, and increase public health risks [3]. Developing countries experience more severe pressures due to insufficient wastewater treatment infrastructure and uncontrolled pollutant discharge into river systems [4]. Sustainable water management has therefore become a global priority under the framework of Sustainable Development Goal 6 (SDG 6), which emphasizes universal access to clean water and improved water quality through pollution reduction and integrated water resource management [5].

The severity and form of river pollution are also influenced by seasonal hydrological conditions. Seasonal variability strongly influences river water quality dynamics, particularly in tropical regions where precipitation patterns regulate pollutant transport, sediment mobility, and dilution processes [6, 7]. Wet-season conditions generally increase surface runoff and nutrient transport from agricultural and urban landscapes, whereas dry-season conditions often intensify pollutant accumulation because of reduced river discharge, lower dilution capacity, and prolonged water residence time. Seasonal monitoring is essential for identifying temporal variations in physicochemical water-quality parameters and improving understanding of river-system responses to changing hydrological conditions, including fluctuations in river discharge, runoff, and precipitation patterns. Water-quality assessment incorporating seasonal variability provides essential scientific evidence for environmental protection, adaptive water-resources management, and sustainable decision-making under changing hydrological and anthropogenic conditions [2, 8]. Reliable seasonal evaluation also supports policy formulation for pollution control and ecosystem-restoration strategies in highly stressed watersheds.

A comprehensive assessment of seasonal river-water conditions requires an approach that integrates multiple physicochemical variables into an interpretable indicator. The Water Quality Index (WQI) is widely applied to integrate measured water-quality parameters into a single numerical value that represents the overall status of a water body relative to established environmental standards [9, 10]. In this study, WQI was used as an assessment and communication tool to classify the observed water-quality condition of the monitored Brantas River reaches. It was not used to identify, quantify, or prove the specific causes of pollution. Interpretation of

potential pollution pressures was based on the spatial pattern of observed parameters, field-based contextual information, and surrounding land-use characteristics.

Geographic Information System (GIS)-based analysis complements WQI assessment by visualizing the spatial distribution of station-based water-quality conditions and identifying river reaches that require closer attention [11, 12]. GIS interpolation does not directly measure pollutant loads or establish causal relationships between land use and water-quality deterioration. Instead, it provides a spatial representation of observed WQI values and supports interpretation of possible environmental pressures in relation to urban settlements, agricultural land, and industrial areas. This distinction is important because pollutant-source attribution requires direct discharge measurements, tributary monitoring, or more comprehensive hydrological and land-use analyses.

These environmental concerns are particularly relevant to the Brantas River, one of the most important river systems in Indonesia. The river supports domestic water supply, irrigation, fisheries, industrial activities, and hydropower generation in East Java Province. Rapid urban development, land-use change, and economic growth throughout the Brantas watershed have intensified pollutant inputs originating from domestic wastewater, agricultural runoff, livestock activities, and industrial effluents, thereby increasing environmental pressure on river-water quality [13, 14]. Several studies reported declining water quality conditions in parts of the Brantas River due to elevated concentrations of organic pollutants, nutrients, and suspended solids [15, 16]. Environmental pressures within the watershed are expected to intensify because of increasing population density and land-use conversion in urban and peri-urban areas [17]. These conditions indicate the urgent need for integrated environmental monitoring and spatial assessment to support sustainable river-management initiatives.

Previous studies have widely applied seasonal water-quality assessment, WQI, GIS mapping, and multivariate statistical analysis in river systems across different climatic and land-use settings [18]. The combination of WQI and GIS should not be regarded as a methodological novelty by itself [19]. Its value lies in the ability to organize station-based observations into an accessible representation of water-quality status and spatial variation that can support environmental monitoring and management decisions. The academic contribution of the present study is a seasonally comparative assessment of selected mainstem reaches of the Brantas River using a consistent set of physicochemical indicators, Weighted Arithmetic WQI classification, and GIS-based visualization [20]. The study provides a baseline description of dry- and wet-season water-quality conditions across 10 monitoring stations distributed from upstream to downstream river sections [21]. It also identifies river reaches where organic-pollution pressure, sediment loading, and nutrient enrichment require priority attention [22]. The findings are intended to support targeted monitoring and management of the monitored mainstem reaches rather than to characterize all pollution processes across the entire Brantas watershed.

This study also recognizes important spatial limitations. The monitoring design was restricted to 10 stations along the main river channel and did not include systematic sampling of incoming tributaries, point-source discharges, or pollutant loads. The observed spatial patterns should thus be interpreted as a screening-level representation of water-quality conditions along the monitored mainstem, not as direct evidence

of source-specific contributions across the entire watershed. Future studies should integrate tributary monitoring, discharge measurements, land-use quantification, and direct wastewater sampling to strengthen pollution-source attribution.

Recent river-water studies have demonstrated the value of seasonal monitoring, weighted arithmetic water-quality indexing, spatial visualization, and multivariate analysis for describing water-quality variation and supporting management-relevant interpretation [23–27]. Building on these established approaches, this study aimed to: (1) assess seasonal variation in pH, dissolved oxygen, biological oxygen demand, chemical oxygen demand, total suspended solids, nitrate, and phosphate at 10 selected mainstem monitoring stations along the Brantas River, (2) classify the observed water-quality status using the Weighted Arithmetic Water Quality Index; (3) visualize the spatial distribution of station-based WQI conditions using GIS; and (4) examine associations among measured physicochemical parameters to support interpretation of observed water-quality patterns. The analysis was designed as a screening-level assessment of monitored river reaches and was not intended to establish causal pollutant sources or represent all tributary contributions within the Brantas watershed.

2 Materials and methods

2.1 Study area

The Brantas River is a major freshwater system in East Java, Indonesia, supporting domestic water supply, irrigation, fisheries, industry, and hydropower. The river originates in the Mount Arjuno and Mount Kelud region, flows through urban and agricultural areas, and discharges into the Madura Strait. Its main channel extends approximately 320 km [28]. The watershed includes forested uplands, agricultural land, urban settlements, and industrial areas. Its tropical monsoon climate produces distinct dry and wet seasons that influence river discharge, pollutant dilution, runoff, nutrient transport, and sediment loading [29–31]. Ten monitoring stations (S1–S10) were established across upstream, middle-stream, and downstream river sections to represent differences in hydrological conditions, land use, and potential pollution sources. Station selection considered accessibility, environmental representativeness, proximity to urban, agricultural, and industrial activities, and spatial coverage of the watershed. Geographic coordinates were recorded using the Global Positioning System to support GIS-based spatial analysis. The study area and distribution of monitoring stations are shown in Fig. 1.

2.2 Sampling design, sample collection, and laboratory analysis

Water sampling was conducted during two synoptic seasonal field campaigns: a dry-season campaign on 15 August 2025 and a wet-season campaign on 15 January 2026. Ten monitoring stations (S1–S10) were surveyed during each campaign, resulting in 10 station-level observations per season and 20 primary field samples in total. One grab sample was collected from each station during each campaign. Laboratory duplicate measurements were used only to assess analytical precision and were not treated as additional field observations. The monitoring stations were distributed across upstream, middle-stream, and downstream reaches of the Brantas River mainstem. Station selection considered accessibility, environmental representativeness, surrounding land use,

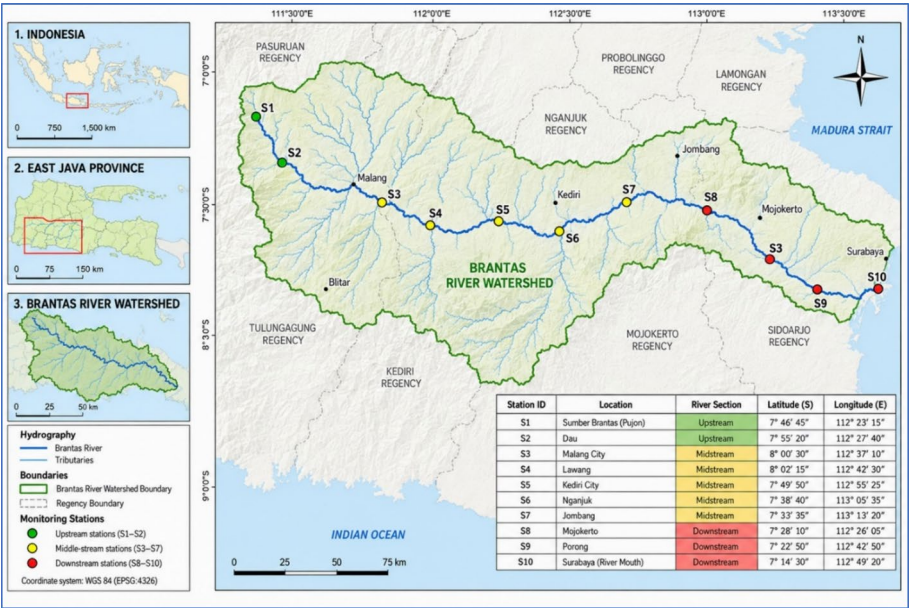


Fig. 1 HYPERLINK "sps:id::fig1||locator:gr1||MediaObject::0" Study area and distribution of 10 monitoring stations (S1–S10) across upstream, middle-stream, and downstream sections of the Brantas River watershed, East Java, Indonesia

population density, agricultural activity, industrial activity, and proximity to potential pollution-pressure areas. This design was intended to provide a screening-level comparison of monitored mainstem reaches. It was not designed to represent all water-quality processes across the entire Brantas watershed because incoming tributaries, point-source outfalls, and pollutant loads were not systematically monitored. Grab samples were collected using pre-cleaned polyethylene bottles at approximately 20–30 cm below the water surface to minimize interference from floating debris and surface contamination. Each bottle was rinsed with river water before sample collection. Geographic coordinates, standardized sampling depth, and identical sample containers were used at all stations to reduce spatial and procedural variation.

Samples designated for laboratory analysis were stored in iceboxes at temperatures below 4 °C and transported promptly to the laboratory at Universitas Negeri Malang, Malang, Indonesia, to minimize chemical and biological alteration during storage [32]. pH and dissolved oxygen (DO) were measured in situ using a portable multiparameter meter to minimize changes associated with sample transportation and exposure. The seven selected parameters were used as a targeted physicochemical screening set for river-water assessment. pH represented acid-base condition, DO represented oxygen availability; BOD and COD represented biodegradable and chemically oxidizable organic pollution; TSS represented suspended particulate and sediment loading; and nitrate and phosphate represented nutrient enrichment. Together, these indicators represent four major water-quality domains: acid-base condition, oxygen–organic matter balance, suspended solids, and nutrient pressure. The parameter set was selected to support compliance-oriented assessment and Weighted Arithmetic Water Quality Index calculation, rather than to provide a complete chemical, microbiological, toxicological, or hydrological characterization of the watershed.

Laboratory analyses followed the procedures specified in Standard Methods for the Examination of Water and Wastewater [32]. DO was determined using the membrane-electrode method, BOD was measured using the five-day incubation method, COD was measured using the dichromate-reflux method, and TSS was determined gravimetrically after filtration and oven drying. Nitrate and phosphate concentrations were measured spectrophotometrically [33, 34]. Potential sampling and analytical errors were minimized through the use of pre-cleaned containers, standardized sampling depth, calibration of analytical instruments before each measurement session, cold preservation during transportation, blank samples, and duplicate laboratory analyses. Standard operating procedures for sample collection, preservation, transportation, and laboratory analysis were consistently applied to support data reliability [35, 36].

The study did not include microbiological indicators, ammoniacal nitrogen, major ions, electrical conductivity, temperature, heavy metals, pesticides, river discharge, or direct wastewater-load measurements. These omissions limit source-specific interpretation and should be considered when interpreting the observed seasonal and spatial patterns. Future monitoring should incorporate these variables together with tributary and outfall sampling to strengthen pollution-source attribution (Table 1).

2.3 Weighted arithmetic water quality index

The Weighted Arithmetic Water Quality Index was applied to summarize the observed status of the monitored Brantas River reaches using seven physicochemical parameters: pH, DO, BOD, COD, TSS, nitrate, and phosphate. The index was used as a compliance-oriented screening tool and was not intended to identify pollutant sources or quantify the relative contribution of domestic, industrial, agricultural, or tributary inputs. The selected parameters were assigned weights using the conventional weighted arithmetic

Table 1 Targeted physicochemical parameters, analytical methods, and rationale for inclusion

Parameter	Unit	Analytical method	Monitoring domain and rationale for inclusion
pH	–	Portable multiparameter meter	Represents acid–base condition and indicates whether river water remains within the applicable regulatory range
Dissolved oxygen (DO)	mg L ^{−1}	Membrane-electrode method	Represents oxygen availability and ecological condition in the river
Biological oxygen demand (BOD)	mg L ^{−1}	Five-day incubation method	Indicates biodegradable organic pollution and oxygen demand associated with microbial decomposition
Chemical oxygen demand (COD)	mg L ^{−1}	Dichromate reflux method	Indicates chemically oxidizable organic and inorganic matter
Total suspended solids (TSS)	mg L ^{−1}	Gravimetric method	Represents suspended particulate matter, sediment transport, and potential runoff-related loading
Nitrate	mg L ^{−1}	Spectrophotometry	Represents nutrient enrichment potentially associated with agricultural runoff, domestic wastewater, and organic decomposition
Phosphate	mg L ^{−1}	Spectrophotometry	Represents nutrient enrichment and potential eutrophication pressure

approach, in which the relative weight of each parameter is calculated from its applicable permissible standard value. The weighting system is thus regulatory-standard based rather than expert-assigned or locally calibrated. Parameters with lower permissible concentrations receive higher relative weights because a smaller exceedance may indicate greater concern relative to the applicable water-quality benchmark.

For the Brantas River assessment, the permissible values used in the WQI calculation were based on the Indonesian surface-water-quality standards adopted for this study under Government Regulation No. 22 of 2021, Annex VI, Class II. The use of this standardized weighting approach was considered appropriate because the study aimed to compare observed concentrations consistently against nationally applicable regulatory thresholds. No Brantas-specific adjustment of WQI weights was applied because no validated local weighting system based on pollutant toxicity, ecological sensitivity, water use, discharge, or exposure characteristics was available. The resulting WQI values should consequently be interpreted as standardized regulatory-screening indicators rather than locally calibrated ecological-risk scores.

The Weighted Arithmetic Water Quality Index (WQI) was used to integrate pH, dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), total suspended solids (TSS), nitrate, and phosphate into a single indicator of river-water quality status. The method assigns a unit weight and quality rating to each parameter according to its applicable water-quality standard [37, 38].

The unit weight of each parameter was calculated as:

$$W_i = \frac{K}{S_i}$$

where (W_i) is the relative weight of parameter (i), (S_i) is the corresponding permissible standard value, and (K) is the proportionality constant calculated as:

$$K = \frac{1}{\sum_{i=1}^n \left(\frac{1}{S_i} \right)}$$

The quality rating for each parameter was calculated using:

$$Q_i = 100 \left(\frac{V_i - V_{i0}}{S_i - V_{i0}} \right)$$

where (Q_i) is the quality rating, (V_i) is the measured value of parameter (i), (V_{i0}) is the ideal value, and (S_i) is the applicable standard value. The ideal values were set at 7.0 for pH, 14.6 mg L⁻¹ for DO, and zero for BOD, COD, TSS, nitrate, and phosphate. The standard values used in the calculation were 9 for pH, 4 mg L⁻¹ for DO, 3 mg L⁻¹ for BOD, 25 mg L⁻¹ for COD, 50 mg L⁻¹ for TSS, 10 mg L⁻¹ for nitrate, and 0.2 mg L⁻¹ for phosphate.

The final WQI value was calculated using:

$$WQI = \frac{\sum_{i=1}^n Q_i W_i}{\sum_{i=1}^n W_i}$$

where n is the total number of parameters included in the calculation ($n = 7$). WQI values were calculated separately for each monitoring station during the dry- and wet-season sampling campaigns. Lower WQI values indicate better water-quality conditions, whereas higher values indicate increasing pollution intensity and environmental degradation [39, 40]. The calculated WQI values were classified as excellent, good, poor, very poor, or unsuitable according to the categories presented in Table 2.

2.4 GIS-based spatial analysis

GIS-based spatial analysis was used to visualize the distribution of water-quality conditions and identify potential pollution hotspots along the Brantas River. Spatial interpolation supports interpretation of the relationships among land use, pollution intensity, and ecological vulnerability within a watershed [41, 42]. Geographic coordinates recorded at each sampling station using the Global Positioning System were imported into ArcGIS to establish the spatial database. Seasonal values of pH, dissolved oxygen, biological oxygen demand, chemical oxygen demand, total suspended solids, nitrate, phosphate, and the Weighted Arithmetic Water Quality Index were integrated into the GIS environment for mapping and spatial interpretation [11].

The Inverse Distance Weighting interpolation method was applied to estimate water-quality patterns along unsampled river segments. This method assumes that observations from stations closer to a predicted location have a greater influence than those located farther away [43, 44]. The resulting spatial surfaces were classified according to WQI categories to identify river segments with relatively good, poor, or very poor water-quality conditions. Spatial patterns were interpreted together with the surrounding land-use characteristics, including urban settlements, industrial areas, agricultural land, and densely populated river corridors. This approach supported identification of river sections requiring priority pollution-control measures and more intensive environmental monitoring.

2.5 Correlation analysis and principal component analysis

The unit of analysis was the station-level observation collected during each seasonal campaign. Each season included 10 station observations, resulting in a total of 20 primary observations for the complete dataset. Descriptive statistics were calculated separately for the dry and wet seasons and included sample size, minimum, maximum, mean, and standard deviation for each parameter. Seasonal differences were described using station-level descriptive statistics because the study was designed as a screening-level assessment of monitored mainstem reaches. The same 10 monitoring stations were

Table 2 Classification of the weighted arithmetic water quality index (WQI)

WQI value	Water quality status	Environmental condition
0–25	Excellent	Suitable for drinking and ecological sustainability
26–50	Good	Minor pollution, acceptable for domestic use
51–75	Poor	Polluted water requiring treatment
76–100	Very poor	Significant pollution and ecological stress
> 100	Unsuitable	Severe pollution, unsuitable for use

sampled during both seasonal campaigns, allowing dry- and wet-season values to be compared descriptively across corresponding stations. The interpretation focused on observed seasonal patterns rather than formal watershed-wide statistical inference.

Pearson correlation analysis was used as an exploratory assessment of associations among the measured physicochemical parameters. Correlation coefficients were calculated from the station-season observations, and the results were interpreted as statistical associations rather than evidence of causal relationships or pollution-source contributions. Variables and correlation plots were inspected for extreme outliers and non-linear patterns before interpretation. Principal Component Analysis (PCA) was applied as an exploratory multivariate technique to identify dominant patterns of association among the measured physicochemical parameters. Before PCA, all variables were standardized using z-scores because the parameters were measured in different units and scales. PCA was performed using the correlation matrix of seven variables, namely pH, dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), total suspended solids (TSS), nitrate, and phosphate. Principal components were interpreted based on eigenvalues greater than 1.00, percentage of variance explained, cumulative variance, and component loading values. Loading values with absolute coefficients of 0.50 or higher were considered meaningful for interpretation. PCA was used only to support exploratory interpretation of water-quality patterns and was not used as definitive source-apportionment or causal evidence of pollutant origins. All statistical analyses were performed using IBM SPSS Statistics, while Microsoft Excel was used for data tabulation and graphical presentation. The overall methodological workflow is summarized in Fig. 2.

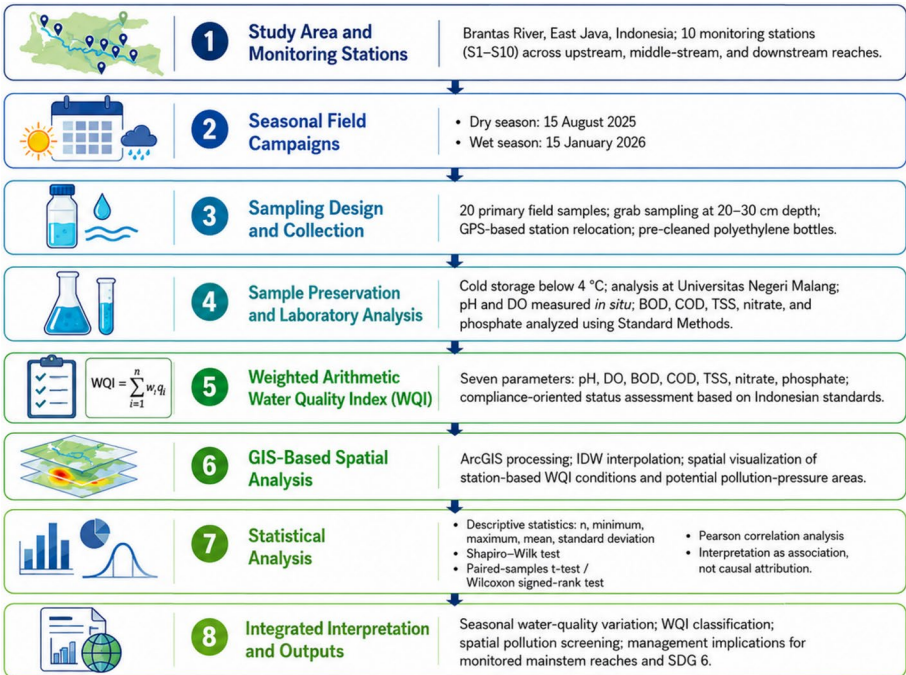


Fig. 2 Methodological workflow for seasonal water-quality assessment and GIS-based pollution mapping of the Brantas River

3 Results and discussion

This section presents the results of seasonal water quality assessment and GIS-based spatial pollution analysis conducted in the Brantas River watershed. Physicochemical parameters measured during dry and wet seasons were evaluated to identify temporal variability, pollution characteristics, and environmental conditions across different monitoring stations. The integration of Water Quality Index (WQI), statistical analysis, and GIS spatial mapping enabled comprehensive interpretation of river pollution dynamics associated with anthropogenic activities and hydrological fluctuations. Seasonal comparison of water quality conditions provides important insights into the influence of rainfall patterns, land use characteristics, and pollutant sources on river ecosystem health. The findings derived from this study are expected to contribute to sustainable watershed management strategies and support Sustainable Development Goal 6 (SDG 6) related to improving water quality and reducing environmental pollution.

3.1 Seasonal water quality variation

Seasonal differences were observed in the physicochemical characteristics of the monitored Brantas River stations during the study period. Differences in rainfall intensity, river discharge, surface runoff, and anthropogenic activities contributed to fluctuations in water quality parameters between dry and wet seasons. Seasonal assessment is important in tropical river systems because hydrological changes strongly affect pollutant transport, nutrient dynamics, dilution processes, and organic matter accumulation [45]. The measured parameters including pH, dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), total suspended solids (TSS), nitrate, and phosphate exhibited noticeable spatial-temporal variability across monitoring stations. Descriptive statistics were calculated separately for the dry and wet seasons based on 10 station-level observations in each season. The values presented include sample size (*n*), minimum, maximum, mean, and standard deviation for each physicochemical parameter. Therefore, the results should be interpreted as a screening-level seasonal assessment of the monitored Brantas River mainstem stations rather than as a complete statistical representation of the entire Brantas watershed.

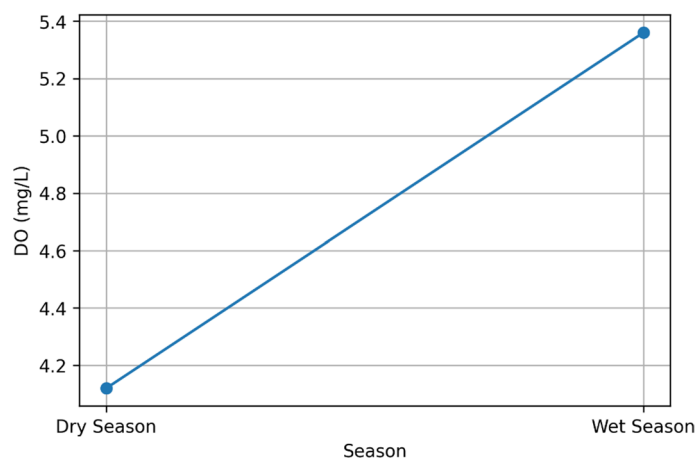
Table 3 provides a complete descriptive summary of seasonal water-quality variation across the monitored stations. The minimum and maximum values show the actual observed range across the 10 monitoring stations in each season, while the mean and standard deviation summarize the central tendency and dispersion of the station-level observations. pH remained within the applicable standard in both seasons, whereas DO was close to the minimum threshold during the dry season. BOD exceeded the applicable standard in both dry and wet seasons, indicating persistent organic-pollution pressure. TSS and phosphate also exceeded the standards, with higher values during the wet season, suggesting increased sediment and nutrient transport during periods of greater runoff.

The seasonal values presented in Table 3 show distinct differences in water-quality conditions between dry- and wet-season sampling. Mean pH values were 7.42 ± 0.31 during the dry season and 7.18 ± 0.27 during the wet season, both within the applicable range of 6–9. Dissolved oxygen (DO) increased from $4.12 \pm 0.84 \text{ mg L}^{-1}$ in the

Table 3 Descriptive statistics of physicochemical parameters in the Brantas River during dry and wet seasons

Parameter	Unit	Season	n	Min	Max	Mean $\pm$ SD	Indonesian water quality standard
pH	pH unit	Dry	10	7.11	7.73	7.42 $\pm$ 0.31	6–9
pH	pH unit	Wet	10	6.91	7.45	7.18 $\pm$ 0.27	6–9
Dissolved oxygen (DO)	mg L ⁻¹	Dry	10	3.28	4.96	4.12 $\pm$ 0.84	≥ 4
Dissolved oxygen (DO)	mg L ⁻¹	Wet	10	4.57	6.15	5.36 $\pm$ 0.79	≥ 4
Biological oxygen demand (BOD)	mg L ⁻¹	Dry	10	5.12	8.58	6.85 $\pm$ 1.73	≤ 3
Biological oxygen demand (BOD)	mg L ⁻¹	Wet	10	3.51	6.33	4.92 $\pm$ 1.41	≤ 3
Chemical oxygen demand (COD)	mg L ⁻¹	Dry	10	19.19	30.15	24.67 $\pm$ 5.48	≤ 25
Chemical oxygen demand (COD)	mg L ⁻¹	Wet	10	14.07	22.61	18.34 $\pm$ 4.27	≤ 25
Total suspended solids (TSS)	mg L ⁻¹	Dry	10	53.16	83.68	68.42 $\pm$ 15.26	≤ 50
Total suspended solids (TSS)	mg L ⁻¹	Wet	10	74.33	115.09	94.71 $\pm$ 20.38	≤ 50
Nitrate (NO ₃ ⁻)	mg L ⁻¹	Dry	10	2.17	3.51	2.84 $\pm$ 0.67	≤ 10
Nitrate (NO ₃ ⁻)	mg L ⁻¹	Wet	10	3.09	4.73	3.91 $\pm$ 0.82	≤ 10
Phosphate (PO ₄ ³⁻)	mg L ⁻¹	Dry	10	0.37	0.59	0.48 $\pm$ 0.11	≤ 0.2
Phosphate (PO ₄ ³⁻)	mg L ⁻¹	Wet	10	0.57	0.89	0.73 $\pm$ 0.16	≤ 0.2

Minimum and maximum values represent the actual observed range across the 10 monitoring stations in each season. Mean and standard deviation were calculated from the same station-level observations

**Fig. 3** Seasonal variation of dissolved oxygen (DO) concentrations

dry season to 5.36 $\pm$ 0.79 mg L⁻¹ in the wet season. Although both values met the minimum standard of 4 mg L⁻¹, the dry-season mean was only 0.12 mg L⁻¹ above this threshold, indicating limited oxygen availability under low-flow conditions. This pattern agrees with recent evidence indicating that low-flow conditions reduce stream reaeration and enhance microbial oxygen consumption associated with persistent organic matter inputs, leading to lower dissolved oxygen concentrations during dry periods [2, 46] (Fig. 3).

BOD reached 6.85 $\pm$ 1.73 mg L⁻¹ during the dry season and 4.92 $\pm$ 1.41 mg L⁻¹ during the wet season. These concentrations exceeded the applicable limit of 3 mg L⁻¹ by 128% and 64%, respectively. COD was 24.67 $\pm$ 5.48 mg L⁻¹ during the dry season,

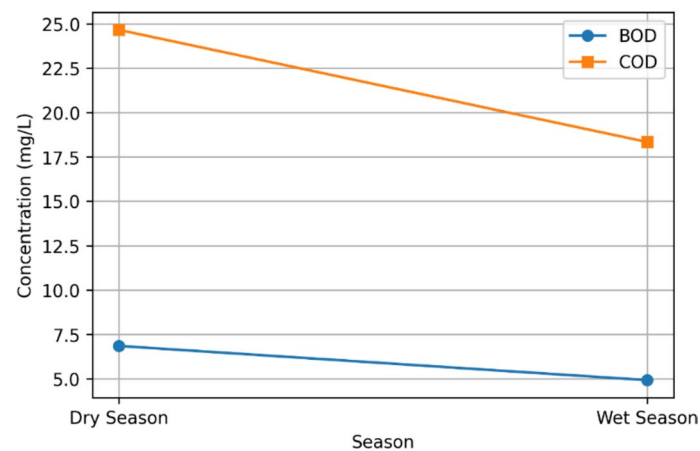


Fig. 4 Seasonal variation of BOD and COD concentrations

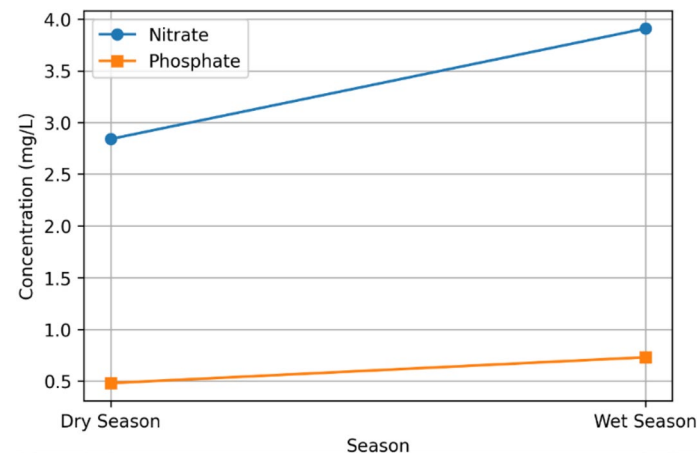


Fig. 5 Seasonal variation of nitrate and phosphate concentrations

close to the applicable limit of 25 mg L^{-1} , and declined to $18.34 \pm 4.27 \text{ mg L}^{-1}$ during the wet season. The concurrent dry-season occurrence of lower DO, elevated BOD, and higher COD indicates increased organic and oxidizable pollutant pressure under reduced dilution conditions. This result agrees with the reported condition of the Brantas watershed, where domestic, industrial, and agricultural activities have been associated with exceedances of BOD, DO, and TSS thresholds [15] (Fig. 4).

TSS showed the strongest wet-season increase, rising from $68.42 \pm 15.26 \text{ mg L}^{-1}$ in the dry season to $94.71 \pm 20.38 \text{ mg L}^{-1}$ in the wet season. Both concentrations exceeded the applicable limit of 50 mg L^{-1} , by 36.8% during the dry season and 89.4% during the wet season. Nitrate remained below the applicable limit of 10 mg L^{-1} in both seasons, with concentrations of $2.84 \pm 0.67 \text{ mg L}^{-1}$ during the dry season and $3.91 \pm 0.82 \text{ mg L}^{-1}$ during the wet season. In contrast, phosphate exceeded the limit of 0.2 mg L^{-1} in both periods, reaching $0.48 \pm 0.11 \text{ mg L}^{-1}$ during the dry season and $0.73 \pm 0.16 \text{ mg L}^{-1}$ during the wet season. The wet-season rise in TSS, nitrate,

and phosphate indicates increased sediment and nutrient transport from agricultural land, urban surfaces, and riverbank erosion during periods of higher runoff (Fig. 5).

The seasonal pattern observed in the Brantas River is comparable with recent studies in tropical and Indonesian river systems, where dry-season conditions commonly intensify organic pollution through reduced dilution, whereas wet-season conditions increase suspended solids and nutrient loading through runoff processes [24, 31]. The present findings indicate two distinct management concerns: control of persistent organic wastewater inputs during low-flow conditions and reduction of sediment and nutrient delivery during the wet season. These results provide a quantitative basis for identifying priority management concerns within the monitored Brantas River mainstem reaches, particularly dry-season organic-pollution control and wet-season sediment–nutrient runoff reduction.

3.2 Water quality index (WQI) assessment

The Weighted Arithmetic Water Quality Index values ranged from 42 to 95 across the monitoring observations, indicating water-quality conditions from good to very poor. Values between 42 and 50 were classified as good, values from 51 to 75 as poor, and values from 76 to 95 as very poor. This range indicates that the Brantas River did not show uniform water-quality conditions; relatively better conditions occurred in less disturbed upstream sections, whereas higher WQI values were associated with middle-stream and downstream areas exposed to cumulative domestic wastewater, industrial activity, agricultural runoff, and suspended-solid inputs.

The WQI pattern is supported by the parameter-level results in Table 3. Elevated BOD, TSS, and phosphate were the main contributors to higher WQI values because these parameters exceeded their applicable standards during one or both sampling periods. The dry-season BOD mean of 6.85 mg L^{-1} exceeded the standard by 128%, whereas wet-season TSS and phosphate reached 94.71 mg L^{-1} and 0.73 mg L^{-1} , exceeding their standards by 89.4% and 265%, respectively. These results explain why water-quality status deteriorated in locations affected by combined organic, sediment, and nutrient pollution pressures.

The WQI range observed in the Brantas River was less severe than the range reported for the Lelliso River, Ethiopia, where Weighted Arithmetic WQI values ranged from 79.6 to 197 during the dry season and from 223.8 to 424.4 during the rainy season [24]. Direct numerical comparison should be interpreted cautiously because the number of parameters, water-quality standards, and intended water uses differed between studies. Nevertheless, both studies demonstrate that WQI can identify substantial spatial and seasonal deterioration in rivers influenced by urban activities, agriculture, and inadequate waste management.

The classification of Water Quality Index values across monitoring stations is presented in Fig. 6. The figure demonstrates clear spatial differences in river pollution status, with upstream areas classified as good to moderately polluted conditions and downstream sections categorized as poor to very poor conditions. The increasing WQI trend from upstream to downstream areas indicates cumulative pollutant accumulation along the river flow direction. Spatial deterioration patterns consistent with the present findings have been widely reported in river systems experiencing

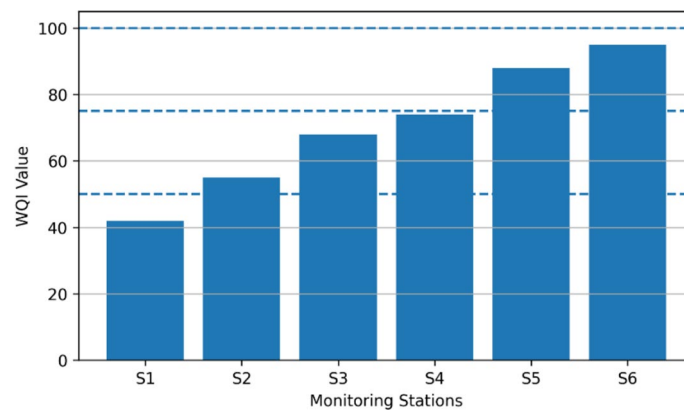


Fig. 6 Water Quality Index classification across monitoring stations

rapid urbanization, increasing anthropogenic pressures, and inadequate wastewater management, resulting in progressive downstream degradation of water quality [2]. The spatial classification derived from WQI assessment provides important scientific evidence for identifying environmentally vulnerable river segments requiring priority management intervention. The classification of Water Quality Index values across monitoring stations is presented in Fig. 6.

The WQI assessment results indicate that several sections of the Brantas River are experiencing moderate to severe pollution pressures associated with anthropogenic activities and watershed land-use changes. Water quality deterioration observed in middle-stream and downstream areas may threaten aquatic ecosystem health, freshwater availability, and the sustainability of river ecosystem services for surrounding communities. Elevated WQI values also indicate increased ecological risks associated with eutrophication, dissolved oxygen depletion, habitat degradation, and declining ecosystem functions in polluted river reaches [2, 8]. Effective pollution control strategies focusing on domestic wastewater management, industrial effluent treatment, and sustainable agricultural practices are essential for improving river water quality conditions. Strengthening environmental monitoring programs and watershed governance is also necessary to support long-term river sustainability and SDG 6 implementation in the Brantas River basin.

The integration of Water Quality Index assessment with seasonal monitoring and GIS spatial analysis provides a comprehensive framework for understanding pollution dynamics and environmental vulnerability within tropical river systems. WQI analysis improves interpretation of spatial-temporal pollution variability and supports evidence-based environmental management decision-making processes [38]. The findings obtained in this study may assist local governments and environmental agencies in prioritizing river restoration programs and pollution mitigation strategies within the Brantas watershed. The applied assessment framework may also serve as a practical reference for future water quality investigations in other tropical watersheds experiencing comparable environmental challenges and anthropogenic pressures.

3.3 Spatial pollution distribution

GIS-based spatial analysis was conducted to visualize the distribution of river water quality conditions along the Brantas River using Water Quality Index (WQI) values. Spatial interpolation analysis revealed a clear upstream-to-downstream gradient of pollution intensity within the watershed. Upstream sections generally exhibited lower WQI values associated with better environmental conditions, while downstream areas showed significantly higher pollution levels due to cumulative anthropogenic pressures. Spatial deterioration patterns consistent with the present findings have been reported in tropical river systems influenced by rapid urbanization and industrial development [47]. Spatial mapping provides important scientific information for identifying environmentally vulnerable river segments and supporting sustainable watershed management strategies.

The GIS interpolation results demonstrated that upstream areas located near forested and less populated regions were classified within good to moderate water quality categories. Middle-stream areas exhibited moderate to poor conditions due to increasing agricultural runoff and urban settlement activities, while downstream regions near Mojokerto and Surabaya showed poor to very poor WQI classifications associated with intensive industrial and domestic wastewater discharge. Increasing pollution intensity along the river flow direction indicates cumulative pollutant accumulation within the watershed system. Environmental distribution patterns consistent with the present findings were documented in previous environmental studies evaluating anthropogenic impacts on river ecosystems in developing regions [48]. The interpolated WQI map in Fig. 7 represents the mean spatial pattern derived from dry- and wet-season station values. The mapped surface should be interpreted as a spatial representation of station-based observations rather than as direct measurements at all unsampled river locations.

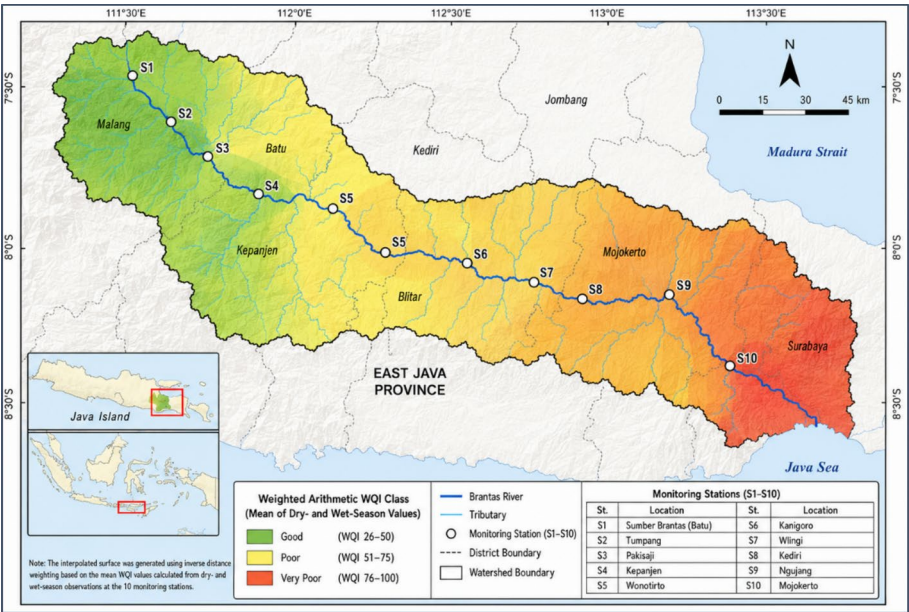


Fig. 7 Spatial distribution of mean Weighted Arithmetic Water Quality Index (WQI) values across 10 monitoring stations in the Brantas River. WQI values were calculated separately for the dry- and wet-season campaigns and averaged for GIS interpolation. WQI classes indicate good (26–50), poor (51–75), and very poor (76–100) water-quality conditions

The spatial distribution map highlights several pollution hotspot areas located in middle-stream and downstream sections of the Brantas River, particularly in regions characterized by high population density, industrial development, and intensive agricultural activities. Elevated WQI values observed in these areas indicate substantial environmental stress associated with organic pollution, nutrient enrichment, and suspended solid accumulation. Industrial zones and densely populated settlements contribute significant domestic and industrial wastewater discharge into river channels, resulting in increased biological oxygen demand and chemical oxygen demand concentrations [11]. Agricultural land surrounding the watershed also contributes substantial nitrate and phosphate inputs through fertilizer runoff during rainfall events, particularly during the wet season, when enhanced surface runoff increases nutrient transport from cultivated areas into river systems [49, 50]. These combined anthropogenic pressures accelerate deterioration of river water quality and increase ecological vulnerability in downstream river segments.

The GIS-based pollution distribution also demonstrates the influence of watershed land use patterns on river ecosystem conditions. Upstream areas with greater vegetation cover and lower anthropogenic disturbance generally exhibited better water quality conditions due to reduced pollutant loading and higher natural filtration capacity. In contrast, downstream urbanized areas showed severe pollution intensity caused by cumulative contaminant transport from upstream sections combined with local wastewater discharge sources. Relationships between land use change and river pollution distribution consistent with the present findings have been widely documented in tropical watersheds undergoing rapid socio-economic development [41]. Spatial analysis therefore provides valuable insights into the interaction between environmental degradation and watershed management practices within the Brantas River basin.

The identified spatial pollution patterns indicate the urgent need for integrated river management strategies focusing on pollution control, wastewater treatment improvement, and sustainable land use planning. Priority management interventions should be directed toward downstream regions experiencing poor to very poor WQI conditions to minimize ecological degradation and protect freshwater resources for surrounding communities. Strengthening environmental regulations related to industrial effluent discharge and domestic wastewater management is also necessary to reduce pollutant loading entering the river system. Sustainable agricultural practices such as controlled fertilizer application and riparian vegetation restoration may contribute to reducing nutrient transport and sediment runoff into surface water bodies [48]. Effective watershed governance integrating scientific monitoring and spatial environmental assessment is therefore essential for improving long-term river sustainability.

The integration of GIS spatial analysis with Water Quality Index assessment proved effective for identifying pollution hotspots and evaluating environmental vulnerability across the Brantas watershed. Spatial visualization of pollution intensity improves interpretation of environmental conditions and supports evidence-based policymaking for river rehabilitation and sustainable water resource management. GIS-generated pollution maps also facilitate communication of environmental risks to stakeholders and local governments responsible for watershed management. The findings derived from this study contribute important scientific information for supporting Sustainable

Development Goal 6 (SDG 6) related to clean water availability, pollution reduction, and sustainable watershed protection. The applied GIS-WQI framework may also serve as a practical reference for future river pollution assessment studies in other tropical developing regions experiencing comparable environmental pressures.

3.4 Pollution sources and environmental implications

Anthropogenic activities within the Brantas watershed were identified as the primary contributors to river water quality degradation observed during the monitoring period. Rapid urbanization, industrial expansion, agricultural intensification, and increasing population density have significantly altered environmental conditions throughout the watershed. Human activities surrounding river corridors generate substantial pollutant inputs that directly affect physicochemical characteristics and ecological health of aquatic ecosystems [51]. Spatial analysis conducted in this study demonstrated that pollution intensity increased considerably in middle-stream and downstream sections where urban settlements and industrial activities are concentrated. These findings indicate that watershed land use patterns strongly influence spatial distribution of river pollution within the Brantas River system.

Domestic wastewater originating from densely populated settlements represents one of the dominant pollution sources contributing to elevated organic pollutant concentrations in the river. Inadequate sanitation infrastructure and untreated household wastewater discharge introduce significant amounts of biodegradable organic matter, nutrients, and suspended solids into river channels [52]. Elevated biological oxygen demand (BOD) and chemical oxygen demand (COD) values observed in downstream areas are closely associated with domestic sewage contamination and urban runoff processes. Environmental conditions consistent with the present findings have been reported in many rapidly developing tropical watersheds where population growth exceeds wastewater treatment capacity [53]. Continuous discharge of untreated domestic wastewater may therefore accelerate ecological degradation and reduce freshwater quality for surrounding communities.

Agricultural activities within the Brantas watershed also contribute substantially to nutrient enrichment and sediment transport in the river system. Intensive agricultural practices involving excessive fertilizer application increase nitrate and phosphate loading through surface runoff, particularly during wet season conditions. Livestock farming activities located near riverbanks may also contribute organic waste and microbial contamination into surface water bodies. Elevated nutrient concentrations observed in several monitoring stations indicate increasing eutrophication risk associated with agricultural runoff and land use conversion. Nutrient pollution patterns observed in the present study are consistent with recent findings from tropical and subtropical river systems, where agricultural intensification, land-use change, and inadequate watershed management have substantially increased nitrogen and phosphorus inputs, accelerating river-water quality degradation [2, 8]. Long-term nutrient enrichment may stimulate excessive algal growth and reduce dissolved oxygen availability within aquatic ecosystems.

Industrial activities distributed along the middle-stream and downstream sections of the Brantas River represent another major source of environmental pollution. Industrial

sectors including textile manufacturing, food processing, chemical industries, and paper production contribute wastewater containing organic pollutants, suspended solids, and chemical contaminants into river systems [54]. Several industrial areas located near Mojokerto and Surabaya exhibited high Water Quality Index (WQI) values indicating poor environmental conditions and significant pollution pressure. Industrial wastewater discharge may increase toxicity levels and alter physicochemical balance within freshwater ecosystems when effluent treatment systems are inadequate or poorly managed. Industrial pollution impacts consistent with the present findings have been reported in urban river systems experiencing rapid economic development and industrial expansion [11]. Figure 8 presents land-use patterns and potential anthropogenic pollution-pressure sources identified from field observations and spatial data. These features were used to contextualize the observed WQI distribution and should not be interpreted as direct measurements of pollutant discharge loads.

Environmental implications associated with declining water quality in the Brantas River include ecological degradation, reduced freshwater availability, increased eutrophication risk, and threats to public health. Deteriorating water quality may adversely affect aquatic biodiversity through dissolved oxygen depletion, habitat alteration, and elevated contaminant toxicity, ultimately reducing the ecological integrity and ecosystem services provided by freshwater systems [2, 8]. River pollution also threatens sustainability of water resources used for irrigation, fisheries, and domestic consumption by surrounding communities. The observed environmental degradation therefore highlights the urgent need for integrated watershed management strategies focusing on pollution reduction and sustainable land use planning. Strengthening wastewater treatment infrastructure, improving industrial effluent management, and implementing sustainable agricultural practices are essential measures for minimizing pollutant loading into the river system.

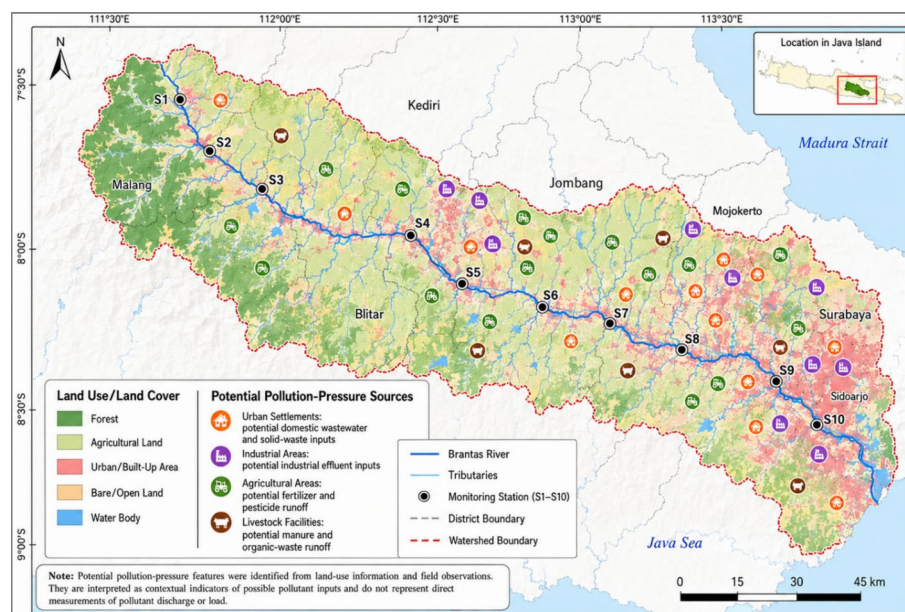


Fig. 8 Spatial distribution of land use/land cover and potential anthropogenic pollution-pressure sources in the Brantas River watershed. Potential pressure sources were identified from land-use information and field observations to support interpretation of the observed WQI pattern and water-quality conditions

The findings derived from this study emphasize the importance of integrating environmental monitoring, spatial analysis, and sustainable watershed governance to support long-term river ecosystem protection. Effective pollution control strategies should involve collaboration among government agencies, local communities, industries, and agricultural sectors to reduce environmental pressures within the Brantas watershed. River restoration programs focusing on riparian vegetation improvement and environmentally sustainable land management practices may contribute to improving water quality conditions and ecological resilience. Integrated environmental management is also critical for achieving Sustainable Development Goal 6 (SDG 6) related to clean water availability, pollution reduction, and sustainable water resource management. The applied spatial assessment framework therefore provides valuable scientific guidance for supporting evidence-based environmental policymaking and watershed rehabilitation initiatives.

3.5 Correlation analysis and principal component analysis

Exploratory correlation analysis was conducted to examine associations among the measured physicochemical parameters in the Brantas River. This analysis was used to support interpretation of the seasonal and spatial water-quality patterns observed from descriptive statistics, WQI classification, and GIS-based mapping [11]. Because the dataset was limited to station-season observations from the monitored mainstem reaches, the correlation results were interpreted as statistical associations rather than as evidence of causal pollution sources.

The lower dry-season DO concentration should not be interpreted as evidence of a direct statistical correlation with BOD or COD because the present seasonal comparison is based on mean values rather than paired station-level correlation results. Nevertheless, the concurrent occurrence of lower DO ($4.12 \pm 0.84 \text{ mg L}^{-1}$) and elevated BOD ($6.85 \pm 1.73 \text{ mg L}^{-1}$) during the dry season indicates increased organic-pollution pressure under low-flow conditions. Reduced discharge may decrease dilution and atmospheric reaeration, while continuous domestic and industrial wastewater inputs can sustain microbial oxygen consumption. Thus, the dry-season pattern is interpreted as the combined influence of reduced hydrological dilution, limited reaeration, and persistent organic loading rather than as a direct one-to-one causal relationship between BOD, COD, and DO.

Pearson correlation analysis revealed strong associations among several physicochemical parameters measured in the Brantas River. Dissolved oxygen (DO) exhibited negative correlations with biological oxygen demand (BOD) and chemical oxygen demand (COD), indicating that reduced oxygen availability coincided with increasing organic pollution levels across the monitored river sections. Similar relationships have been widely reported in river systems experiencing elevated organic loading, where microbial decomposition increases oxygen consumption and contributes to declining dissolved oxygen concentrations [2, 8]. Strong positive associations between BOD and COD indicate that biodegradable and chemically oxidizable pollutants tended to increase together across the observations. Positive associations between nitrate and phosphate also suggest that nutrient enrichment occurred concurrently in several monitored sections. These patterns support the interpretation of organic

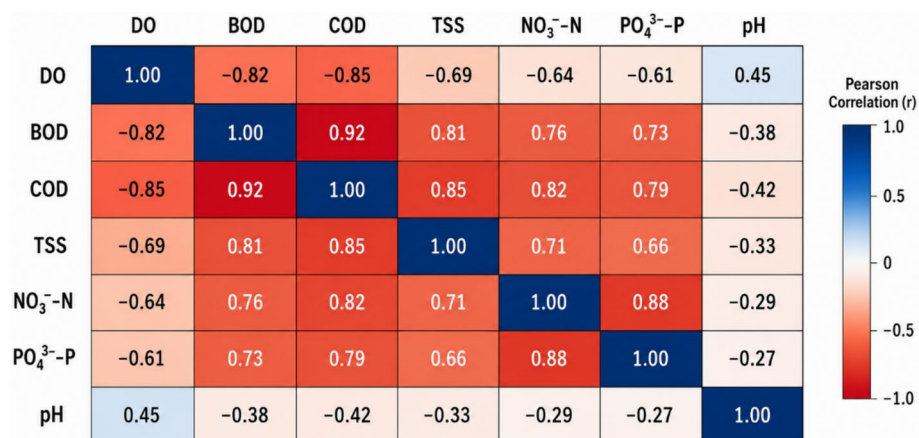


Fig. 9 Pearson correlation matrix of seven physicochemical water-quality parameters across monitoring observations in the Brantas River

Table 4 Eigenvalues and variance explained by principal components

Principal component	Eigenvalue	Variance explained (%)	Cumulative variance (%)	Interpretation
PC1	4.09	58.40	58.40	General pollution and oxygen-depletion gradient
PC2	1.73	24.70	83.10	Runoff-related suspended solid and nutrient gradient
PC3	0.48	6.80	89.90	Not retained
PC4	0.29	4.20	94.10	Not retained
PC5	0.18	2.50	96.60	Not retained
PC6	0.13	1.90	98.50	Not retained
PC7	0.11	1.50	100.00	Not retained

PCA was performed using standardized values of seven physicochemical parameters. Components with eigenvalues greater than 1.00 were retained for interpretation

and nutrient pressure in the Brantas River, however, they should not be interpreted as direct causal evidence or source-apportionment results.

The correlation matrix presented in Fig. 9 demonstrates the interaction patterns among water quality parameters and highlights the dominant influence of organic pollution and nutrient loading within the watershed. Strong correlations among suspended solids, nutrient concentrations, and organic pollution indicators suggest that surface runoff, land-use change, agricultural activities, and urban development collectively contribute to river-water quality degradation. Comparable statistical relationships have been widely reported in river systems experiencing increasing anthropogenic pressures and watershed modification [2, 8]. Correlation analysis therefore provides important scientific evidence regarding the relationships between environmental variables and dominant pollution mechanisms in the Brantas River. The Pearson correlation matrix of physicochemical parameters measured in the Brantas River is presented in Fig. 9.

To strengthen the multivariate interpretation, PCA results are presented in terms of eigenvalues, percentage of variance explained, cumulative variance, and component loadings. The eigenvalue results are shown in Table 4, while the component loading values are presented in Table 5. PCA was used only to support exploratory interpretation of water-quality patterns and was not used as definitive source-apportionment or causal evidence of pollutant origins.

The component loading matrix was used to identify the dominant variables associated with each retained principal component. The loading values are presented in Table 5.

Principal Component Analysis (PCA) was conducted using standardized station-season observations to explore the dominant multivariate patterns among the seven physicochemical parameters. As shown in Table 4, two principal components had eigenvalues greater than 1.00 and were retained for interpretation. PC1 had an eigenvalue of 4.09 and explained 58.40% of the total variance, while PC2 had an eigenvalue of 1.73 and explained 24.70% of the total variance. Together, PC1 and PC2 explained 83.10% of the total variance, indicating that most of the variation in the standardized water-quality dataset could be summarized by these two components.

The loading matrix in Table 5 shows that PC1 was characterized by strong positive loadings for BOD, COD, TSS, nitrate, and phosphate, and a strong negative loading for DO. This pattern indicates that PC1 represents a general pollution gradient associated with organic matter, suspended solids, nutrient enrichment, and reduced oxygen availability. Higher PC1 scores therefore indicate poorer water-quality conditions, whereas lower PC1 scores indicate relatively better oxygen status and lower pollution pressure. PC2 showed meaningful positive loadings for pH and TSS, with moderate contributions from nitrate and phosphate. This component may reflect runoff-related variation and sediment–nutrient transport, particularly during wet-season conditions. The PCA results should be interpreted as exploratory multivariate patterns rather than direct evidence of specific pollutant sources.

The PCA biplot shown in Fig. 10 visualizes the relative associations between monitoring observations and physicochemical variables along the PC1 and PC2 axes. Observations located toward the BOD, COD, TSS, nitrate, and phosphate vectors indicate stronger associations with organic pollution, suspended solids, and nutrient enrichment, whereas observations positioned closer to the DO vector represent relatively better

Table 5 Component loading values of physicochemical parameters on retained principal components

Parameter	PC1	PC2	Dominant interpretation
pH	−0.45	0.73	Acid–base condition
Dissolved oxygen (DO)	−0.88	0.21	Oxygen availability, inversely related to pollution pressure
Biological oxygen demand (BOD)	0.91	−0.18	Biodegradable organic pollution
Chemical oxygen demand (COD)	0.94	−0.11	Chemically oxidizable pollution
Total suspended solids (TSS)	0.76	0.54	Suspended solids and runoff-related loading
Nitrate (NO ₃ [−])	0.82	0.46	Nutrient enrichment
Phosphate (PO ₄ ^{3−})	0.85	0.43	Nutrient enrichment and eutrophication pressure

Loading values with absolute coefficients ≥ 0.50 were considered meaningful for interpretation. Positive and negative signs indicate the direction of association within each principal component and should not be interpreted as independent causal effects

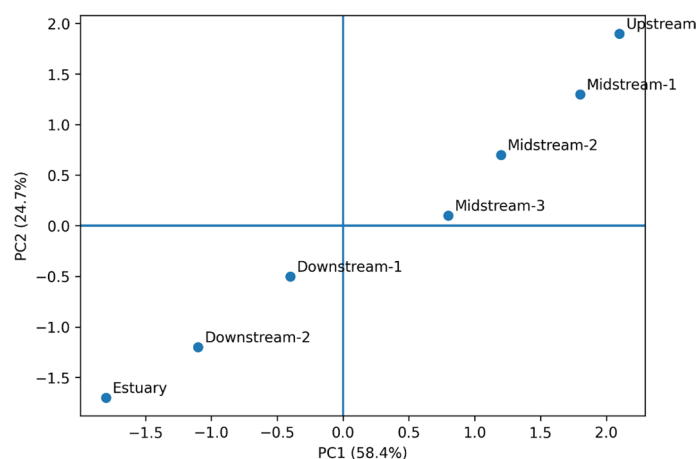


Fig. 10 Principal Component Analysis (PCA) of water quality parameters and monitoring station distribution in the Brantas River

oxygen conditions. This ordination pattern agrees with the WQI and GIS analyses, which identified poorer water-quality conditions in several middle-stream and downstream sections. The PCA biplot should be interpreted as an exploratory multivariate technique that reveals relationships among water-quality variables rather than direct evidence of pollutant-source attribution [2, 55].

PC1 explained 58.40% of the total variance and represents a general pollution gradient associated with BOD, COD, TSS, nitrate, phosphate, and reduced DO. PC2 explained 24.70% of the total variance and reflects runoff-related variation involving pH, suspended solids, and nutrient parameters. The biplot is interpreted as an exploratory visualization of multivariate association rather than direct source-apportionment evidence. The combined correlation and PCA results support the interpretation that organic-pollution indicators, suspended solids, and nutrient parameters tended to vary together across the monitored observations. These statistical patterns are consistent with the seasonal descriptive results and GIS-based WQI maps, which showed dry-season organic-pollution pressure and wet-season sediment–nutrient loading. However, the statistical results should not be interpreted as definitive causal evidence of specific pollutant sources because the study did not include direct discharge measurements, pollutant-load estimation, or systematic tributary monitoring. Instead, correlation and PCA were used to provide supporting evidence for identifying priority monitoring concerns and management needs in the monitored Brantas River mainstem reaches.

3.6 Implications for sustainable water management and SDG 6

The management implications of this study were formulated by linking the seasonal descriptive statistics, WQI classification, GIS-based hotspot distribution, and exploratory PCA results. The evidence indicates that the monitored Brantas River mainstem does not require a uniform management response across all sections. Instead, management priorities should be differentiated by river section and season. Table 3 shows that dry-season conditions were mainly characterized by organic-pollution pressure, particularly elevated BOD and COD combined with relatively low DO. In contrast, wet-season

conditions were characterized by increased TSS, nitrate, and phosphate, indicating stronger sediment and nutrient transport during periods of higher runoff [51]. The GIS-based WQI map in Fig. 7 further shows that poorer water-quality conditions were concentrated mainly in middle-stream and downstream sections, while Fig. 8 provides contextual evidence that these sections are associated with urban settlements, industrial activities, and agricultural land-use pressures. The PCA results in Tables 4, 5, and Fig. 10 support this interpretation by showing that BOD, COD, TSS, nitrate, and phosphate were major contributors to the multivariate pollution gradient, while DO was inversely related to this pollution pattern.

Based on these findings, downstream urban and industrial reaches should be treated as the highest priority for organic-pollution control. This priority is justified by the poor to very poor WQI conditions observed in downstream sections and by the elevated dry-season BOD concentration of $6.85 \pm 1.73 \text{ mg L}^{-1}$, which exceeded the applicable standard of 3 mg L^{-1} . COD also approached the applicable limit during the dry season, reaching $24.67 \pm 5.48 \text{ mg L}^{-1}$. These conditions indicate that low-flow periods require intensified control of organic wastewater inputs. Practical interventions should include upgrading domestic wastewater treatment capacity, expanding decentralized sewage treatment systems in densely populated riverbank settlements, increasing inspection of domestic wastewater discharge points, and strengthening compliance monitoring for industrial effluents in downstream industrial corridors, particularly around urban-industrial sections such as the Mojokerto–Surabaya reach [3]. These measures should be prioritized before and during the dry season, when dilution capacity is reduced and organic-pollution pressure is more evident.

Middle-stream and peri-urban agricultural sections require a different management emphasis. Table 3 shows that wet-season TSS increased from $68.42 \pm 15.26 \text{ mg L}^{-1}$ to $94.71 \pm 20.38 \text{ mg L}^{-1}$, while phosphate increased from $0.48 \pm 0.11 \text{ mg L}^{-1}$ to $0.73 \pm 0.16 \text{ mg L}^{-1}$. Both parameters exceeded the applicable standards, indicating that sediment and nutrient runoff represent major wet-season pressures. Therefore, agricultural management zones should be designated in river sections where agricultural land and runoff pathways are located near the mainstem [54]. Priority measures should include riparian buffer restoration, vegetated filter strips, erosion-control structures, regulated fertilizer timing before heavy rainfall, reduced fertilizer application near riverbanks, and drainage management to reduce sediment and nutrient delivery into the river. These actions are particularly relevant during the wet season, when runoff-driven sediment and nutrient transport increases.

The upstream sections should be maintained as protection and reference zones rather than treated primarily as pollution-control zones. The WQI pattern indicates relatively better conditions in less disturbed upstream areas compared with middle-stream and downstream sections [30]. Management in these areas should focus on preserving vegetation cover, preventing land-use conversion near river corridors, maintaining riparian integrity, and using upstream stations as reference points for long-term monitoring. This distinction is important because the study indicates a downstream deterioration gradient, therefore, maintaining upstream ecological condition can support basin-wide water-quality resilience.

The PCA results should not be interpreted as direct source-apportionment evidence, but they provide useful support for prioritizing management actions. PC1, which explained 58.40% of the variance, was associated with higher BOD, COD, TSS, nitrate, and phosphate and lower DO, indicating a general pollution and oxygen-depletion gradient. PC2, which explained 24.70% of the variance, reflected additional variation related to suspended solids and nutrient transport. These PCA patterns are consistent with the seasonal and spatial results: dry-season organic-pollution pressure and wet-season sediment–nutrient loading. Therefore, the statistical results support the need for two complementary management strategies: organic-wastewater reduction in dry-season and downstream urban-industrial sections, and runoff-control intervention in wet-season and agricultural or peri-urban sections.

From an SDG 6 perspective, the findings provide three specific management directions. First, SDG 6.3 can be supported by reducing untreated wastewater discharge through domestic sewage treatment expansion and stricter industrial effluent compliance in downstream urban-industrial reaches. Second, SDG 6.6 can be supported by protecting riparian ecosystems and reducing sediment and nutrient transport in agricultural and peri-urban sections. Third, SDG 6.5 can be supported by integrating seasonal water-quality monitoring, GIS hotspot mapping, and targeted intervention planning into river-basin management [41]. These recommendations are derived from the observed parameter exceedances, WQI hotspot distribution, and exploratory PCA results rather than from general literature alone.

The practical implication of this study is that Brantas River management should adopt a section-specific and season-specific intervention framework. Downstream urban-industrial sections should be prioritized for domestic wastewater treatment, industrial effluent monitoring, and dry-season organic-pollution control [5]. Middle-stream and agricultural sections should be prioritized for wet-season runoff reduction, riparian restoration, fertilizer management, and sediment-control measures. Upstream sections should be maintained as ecological protection and monitoring reference zones. This targeted strategy is more appropriate than a uniform watershed-wide recommendation because the present study was based on 10 monitored mainstem stations and was designed as a screening-level assessment rather than a complete source-apportionment study of the entire Brantas watershed.

4 Conclusion

This study identified distinct seasonal and spatial water-quality pressures in the Brantas River. Dry-season conditions were characterized by elevated organic-pollution indicators, with BOD reaching $6.85 \pm 1.73 \text{ mg L}^{-1}$, exceeding the applicable standard by 128%, and COD reaching $24.67 \pm 5.48 \text{ mg L}^{-1}$, close to the applicable limit. Wet-season conditions showed higher sediment and nutrient loading, with TSS increasing to $94.71 \pm 20.38 \text{ mg L}^{-1}$ and phosphate reaching $0.73 \pm 0.16 \text{ mg L}^{-1}$, exceeding the applicable standards by 89.4% and 265%, respectively. In contrast, pH and nitrate remained within the applicable limits during both sampling periods. The Weighted Arithmetic Water Quality Index ranged from 42 to 95, indicating conditions from good to very poor across the monitored river sections. GIS-based mapping showed that poorer

water-quality conditions were concentrated in middle-stream and downstream areas exposed to cumulative urban, industrial, and agricultural pressures.

These findings indicate that the Brantas River requires season-specific pollution-control strategies. Domestic wastewater treatment and industrial-effluent compliance should be prioritized during low-flow conditions to reduce organic-pollution pressure, while riparian protection, soil-conservation practices, and more efficient fertilizer management are required to reduce wet-season sediment and nutrient transport. Future research should include repeated multi-year monitoring, river-discharge and rainfall measurements, microbiological indicators, and direct measurements of pollutant loads from domestic, industrial, and agricultural sources. Such information would improve source attribution, strengthen spatial-model validation, and support targeted watershed interventions aligned with SDG 6.

Author contributions

I. J. F. conceptualized the study, conducted field investigation, performed formal analysis, prepared visualization, and wrote the original manuscript draft. H. H. and S. S. supervised the research, contributed to methodology development, and reviewed the manuscript. S. Y. E. contributed to data interpretation, GIS spatial analysis, and manuscript editing. A. W. M. contributed to laboratory analysis, validation, and data curation. L. U. A. contributed to statistical analysis, manuscript review, and final editing. All authors reviewed and approved the final version of the manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study did not involve human participants or animals. Therefore, ethical approval was not required under the institutional guidelines of the participating universities.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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